

Análise de Dados para Business Intelligence

Avaliação Teórico Prática 2

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I. Análise Preditiva de Dados - Regressão Linear.

1. MntVeganVegetarian

- a. I selected the variables **Income**, **Age**, **Education**, and **MntMeatFish**. I chose them because: **Income** and **Recency** directly influence the amount and quantity of customers' spending in the restaurant; **Age**, in my understanding, may affect some dietary restrictions or food preferences; **MntMeatFish** seems to me to have a direct correlation with **MntVeganVegetarian**, as they are opposite entities in the diet.

$$\text{MntVeganVegetarian} = \beta_0 + \beta_1 \times \text{Income} + \beta_2 \times \text{Age} + \beta_3 \times \text{Recency} + \beta_4 \times \text{MntMeatFish}$$

```
Call:
lm(formula = MntVeganVegetarian ~ Income + Age + Recency + MntMeatFish,
    data = sabor)

Residuals:
    Min       1Q   Median       3Q      Max
-13580.1  -1449.0   -325.3    965.0   26750.0

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.176e+03  2.605e+02  -8.352  < 2e-16 ***
Income       7.167e-02  1.930e-03  37.131  < 2e-16 ***
Age         -2.151e+01  4.653e+00  -4.623  3.98e-06 ***
Recency      9.026e-01  1.925e+00   0.469   0.639
MntMeatFish  1.109e-01  2.025e-02   5.475  4.82e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2726 on 2495 degrees of freedom
Multiple R-squared:  0.5058,    Adjusted R-squared:  0.505
F-statistic: 638.5 on 4 and 2495 DF,  p-value: < 2.2e-16
```

- b. **Multiple R-squared: 0.5058.** It indicates that around 50.58% of the variability in the MntVeganVegetarian is explained by the selected variables. This value indicates an **Ajustamento Mediano** and it means that other factors also influence the MntVeganVegetarian.

Adjusted R-squared: 0.505. This value is being adjusted to the number of variables and sample size, confirming the relevance of selected variables for the

model. In this case, Adjusted R-squared is close to Multiple R-squared, indicating that all included predictors contribute meaningfully to the model.

- c. The validity of the model is based on the ANOVA test, with the following hypotheses to be tested:

H0: $R^2_{população} = 0$

Ha: $R^2_{população} \neq 0$

In this case, p-value: $< 2.2e-16$, and **H0 is rejected**. Therefore, **the model is valid** because the independent variable is significant for explaining variations in the dependent variable, indicating that the model as a whole is relevant for predicting MntVeganVegetarian.

- d. **$\beta_0 = -2.176e+03$** $\leftrightarrow$ Indicates that if the client's Income, Age, Recency, and MntMeatFish were hypothetically zero, the estimated MntVeganVegetarian value would be approximately $-2.176e+03$.

$\beta_1 = 7.167e-02$ $\leftrightarrow$ For each unit change in the customer's Income, the MntVeganVegetarian is estimated to increase by $7.167e-02$ units, keeping other variables constant.

$\beta_2 = -2.151e+01$ $\leftrightarrow$ For each unit change in the customer's Age, the MntVeganVegetarian is estimated to decrease by $2.151e+01$ units, keeping other variables constant.

$\beta_3 = 9.026e-01$ $\leftrightarrow$ For each unit change in the customer's Recency, the MntVeganVegetarian is estimated to increase by $9.026e-01$ units, keeping other variables constant.

$\beta_4 = 1.109e-01$ $\leftrightarrow$ For each unit change in the customer's MntMeatFish, the MntVeganVegetarian is estimated to increase by $1.109e-01$ units, keeping other variables constant.

- e. **Income: p-value ≈ 0** , the null hypothesis is rejected, the variable contributes to explaining the variation in the MntVeganVegetarian.

Age: p-value ≈ 0 , the null hypothesis is rejected, the variable contributes to explaining the variation in the MntVeganVegetarian.

Recency: p-value > 0.05 , the null hypothesis is not rejected, and the variable does not contribute to explaining the variation in the MntVeganVegetarian. It is **not statistically significant** in estimating the credit limit and **can be excluded from the model**.

MntMeatFish: p-value ≈ 0 , the null hypothesis is rejected, and the variable contributes to explaining the variation in the MntVeganVegetarian.

f. Pressupostos do modelo.

1. **Linearity of the phenomenon under study.**

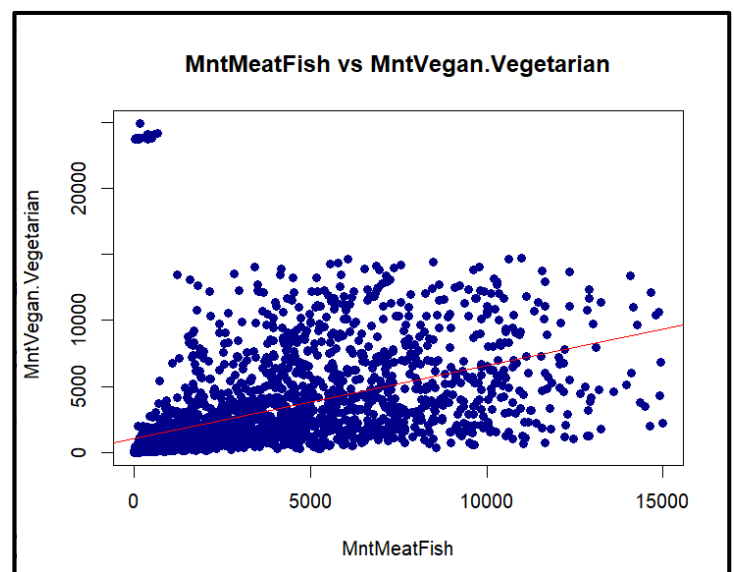
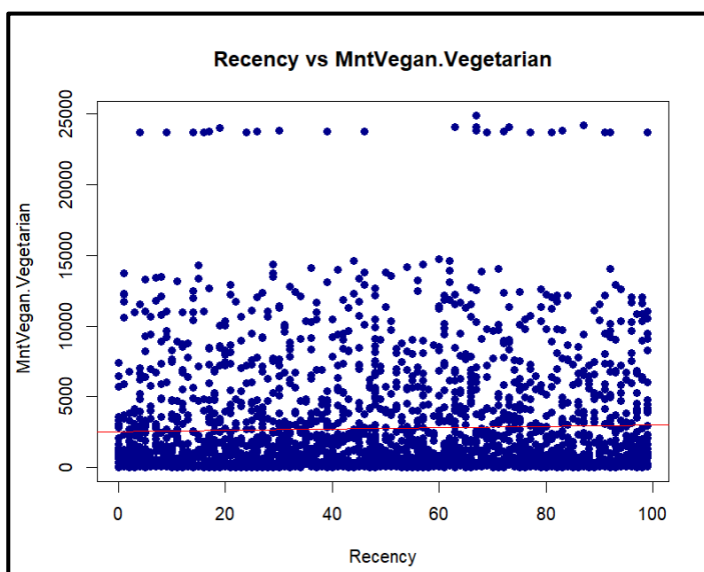
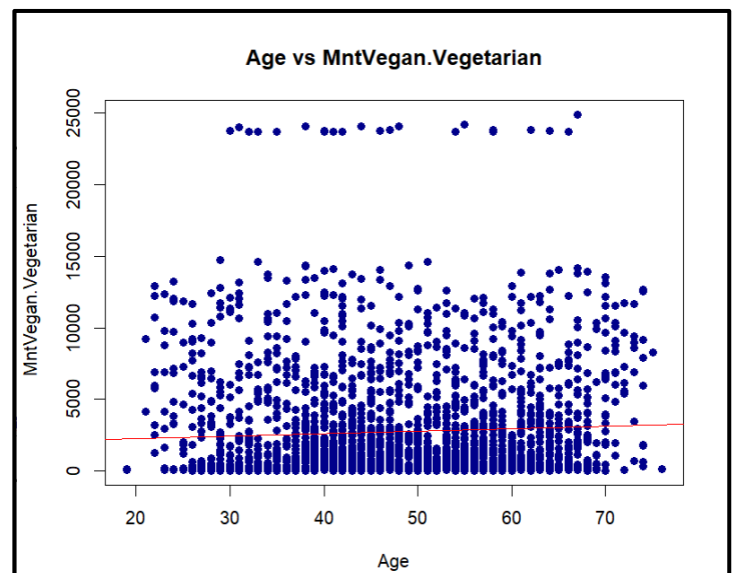
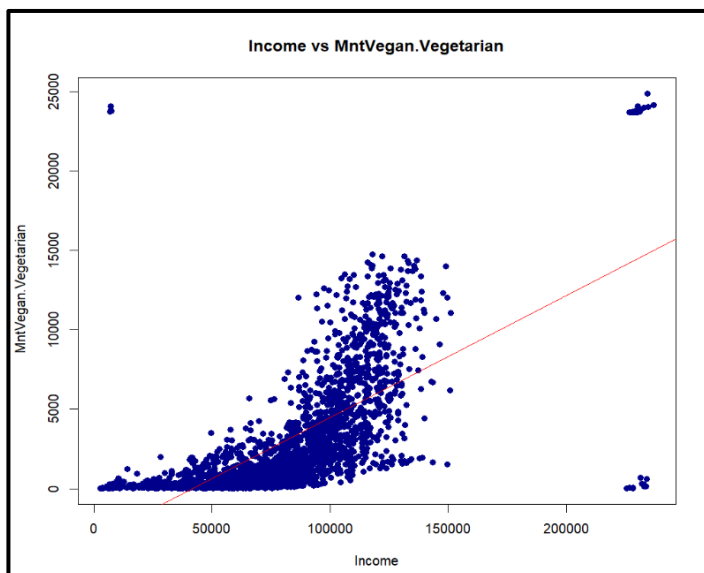
Income: we can see a correlation with MntVeganVegetarian, but it is not really linear.

Age: no correlation with MntVeganVegetarian. Is not significant in explaining the value of MntVeganVegetarian.

Recency: no correlation with MntVeganVegetarian. Is not significant in explaining the value of MntVeganVegetarian.

MntMeatFish: I would say there is no correlation with

MntVeganVegetarian, but based on abline I can conclude that there is a slight correlation.



2. Normality of Errors

H0: errors are following normal distribution;

Ha: errors are not following normal distribution.

p-value ≈ 0 , rejecting H0, therefore errors are not following normal distribution.

The assumption of normality of residuals is not validated.

```
Shapiro-wilk normality test  
data: residuals(model)  
W = 0.83724, p-value < 2.2e-16
```

3. The mean of the random error is zero

p-value ≈ 0 , this indicates that the mean of the residuals is zero and that the model is not systematically over- or under-estimating the true values.

```
> mean(residuals(model))  
[1] -6.313243e-14
```

4. Independence between explanatory variables

The VIF and Tolerance values for Income and MntMeatFish indicate that there are multicollinearity problems. The independent variables are somewhat correlated with each other and therefore it can cause significant problems in the interpretation of the regression coefficients.

The solution to the problem of multicollinearity between the independent variables is:

Increasing the sample size;

Eliminating the variable most correlated with the others;

Focusing on the independent variables

Term	VIF	VIF 95% CI	Increased SE	Tolerance
Income	1.58	[1.50, 1.67]	1.26	0.63
Age	1.03	[1.01, 1.12]	1.02	0.97
Recency	1.00	[1.00, 1.03e+06]	1.00	1.00
MntMeatFish	1.57	[1.49, 1.66]	1.25	0.64

- g. The VIF and Tolerance values for **Income** and **MntMeatFish** indicate that **there are multicollinearity problems**. The independent variables are somewhat correlated with each other and therefore it can cause significant problems in the interpretation of the regression coefficients.

The solution to the problem of multicollinearity between the independent variables is:

Increasing the sample size;

Eliminating the variable most correlated with the others;

Focusing on the independent variables.

- h. With the help of p-values I can conclude that **Gender does not contribute to explaining** the variation in the MntVeganVegetarian. Whereas **Response_Cmp4 is significant** for explaining the variation in the MntVeganVegetarian. This may be because this campaign involved menu items in the Vegan or Vegetarian category. Due to this, participation or non-participation in this campaign is significant for explaining the amount spent on this type of dish.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-2.018e+03	2.690e+02	-7.500	8.84e-14	***
Income	7.028e-02	1.929e-03	36.433	< 2e-16	***
Age	-2.007e+01	4.624e+00	-4.339	1.49e-05	***
Recency	7.528e-01	1.911e+00	0.394	0.69365	
MntMeatFish	6.897e-02	2.124e-02	3.248	0.00118	**
GenderMale	-1.418e+02	1.139e+02	-1.244	0.21347	
Response_Cmp4	1.402e+03	2.268e+02	6.180	7.45e-10	***

II. Análise Descritiva de Dados Multivariados – Análise em Componentes Principais.

1. ACP

a. Appropriateness of the PCA to the data

With a KMO value of 0.86, the PCA's suitability for the data is good, but we can continue with the analysis, the variables under study correlate in the sample.

According to Bartlett's test, we reject the null hypothesis (p-value = 0.0), so we can infer that the variables are correlated in the population.

Kaiser-Meyer-Olkin factor adequacy			\$Schisq
Call: KMO(r = cor_matrix)			
Overall MSA = 0.86			[1] 4564.83
MSA for each item =			
MntMeatFish	MntEntries	MntVeganVegetarian	\$p.value
0.85	0.86	0.85	
MntDrinks	MntDesserts	MntAdditionalRequests	
0.85	0.86	0.87	
			[1] 0

b. Extraction of principal components

	Componente	Valor_Proprio	Variância_Explicada	Variância_Acumulada
1	1	3.1822586	53.037643	53.03764
2	2	0.7300277	12.167129	65.20477
3	3	0.6993844	11.656407	76.86118
4	4	0.4880193	8.133655	84.99483
5	5	0.4588435	7.647391	92.64223
6	6	0.4414665	7.357775	100.00000

1. **Kaiser Method** (Valor_Proprio > 1) → 1 principal component
2. **Variância_Acumulada** (>70%) → 3 principal components
3. **Screeplot** → 4 principal components

2. Using 3 principal components.

- With 3 components extracted, **76,86%** of the total variance is explained (Variância_Acumulada on the image above).

b. Component 1: Side Dishes

- MntEntries

- MntDrinks

- MntDesserts

Component 2: Additional Requests

- MntAdditionalRequests

Component 3: Main Dishes

- MntMeatFish

- MntVeganVegetarian

	RC1	RC3	RC2
MntMeatFish	0.16	0.81	0.37
MntEntries	0.79	0.22	0.16
MntVeganVegetarian	0.41	0.78	-0.07
MntDrinks	0.79	0.21	0.18
MntDesserts	0.75	0.24	0.20
MntAdditionalRequests	0.27	0.14	0.92

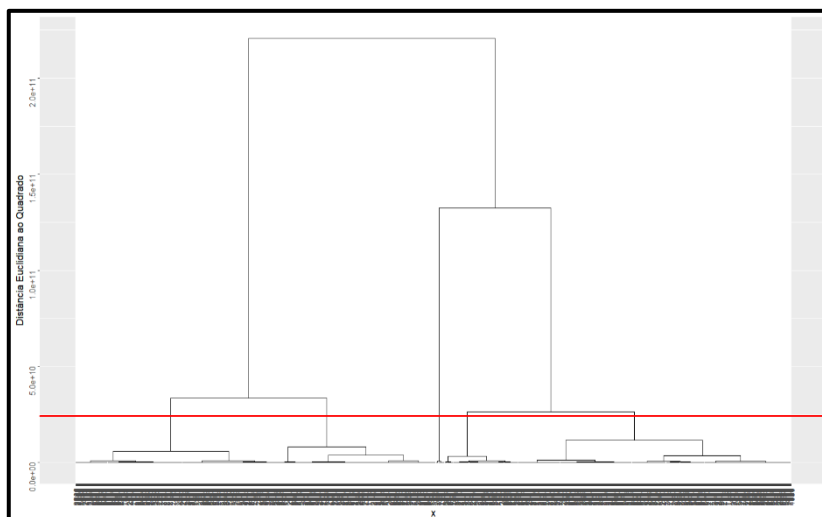
- c. The correlations are high (greater than 0.9 in absolute value), so the indices created represent the scores of the principal components. Therefore we can use them instead of principal components.

```
> print(correlacao_PC1)
[1] 0.9337934
> print(correlacao_PC2)
[1] 0.9168442
> print(correlacao_PC3)
[1] 0.9236438
```

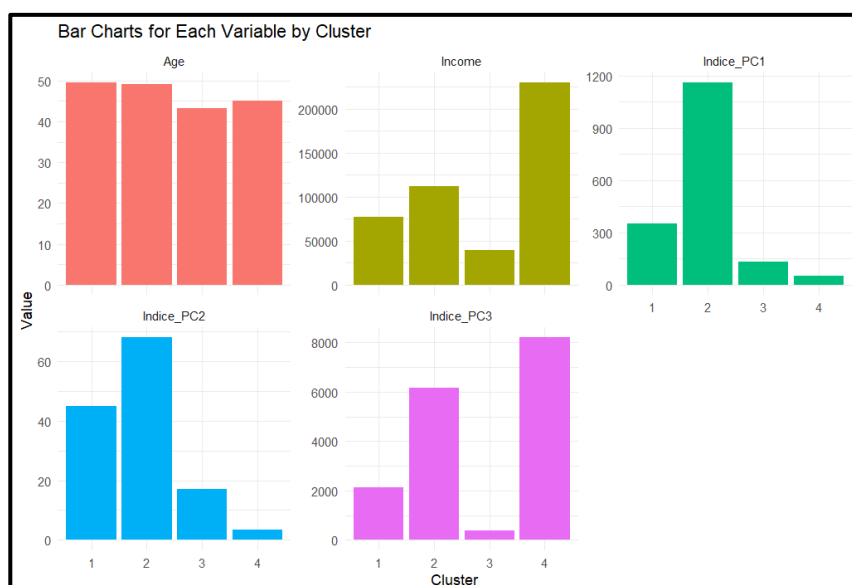
III. Análise Descritiva de Dados Multivariados – Análise de Clusters.

1. Clustering

- a. I believe that the most adequate number of clusters would be five. In my opinion, the choice of more general groups (for example 3) of customers would be incorrect, given our task of creating special promotions, including personalized ones.



C.



	Indice_PC1	Indice_PC2	Indice_PC3	Income	Age
1	352.00767	44.901036	2140.6128	76738.61	49.63867
2	1161.93963	68.091837	6177.4477	111755.97	49.22577
3	134.22263	17.003676	388.0374	39771.79	43.28799
4	51.91398	3.580645	8208.5968	230217.77	45.03226

d. **Cluster 1:**

- Average values for all indicators.
- Moderate spending on main meals and additional requests.

Cluster 2:

- High spending on Indice_PC1 and Indice_PC2 (appetizers, drinks, desserts and additional requests).
- High income.
- Active consumption across all categories.

Cluster 3:

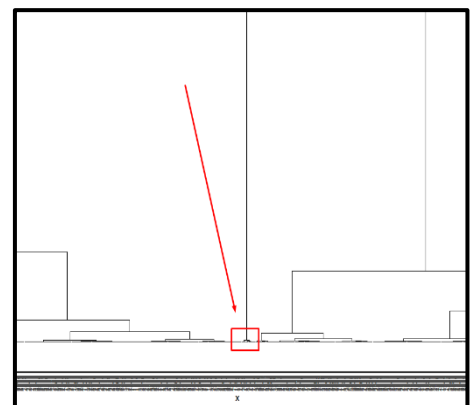
- Low spending in all categories, especially on main meals.
- Lowest income.
- Youngest age of customers.

Cluster 4:

- Highest spending on Indice_PC3 (meat and vegetarian dishes).
- Highest income.
- Minimal spending on appetizers and additional requests.

- e. Among the number of clients entering the clusters, we can notice a clear outlier, it is cluster 4, consisting of only 31 customers, while the number of clients in the other clusters is evenly distributed. This was predictable. These are the clients with the highest income. These are the same outliers we saw in the scatter plot of Income, and they are also the distinct group of people that can be seen in the dendrogram.

	Cluster	Número_de_Casos
1	1	869
2	2	784
3	3	816
4	4	31



f. **Premium main course menu** campaign (Cluster 4)

Campaign Description:

Develop a special premium menu featuring unique meat and vegetarian dishes with gourmet ingredients and signature presentation.

Add chef compliments or small discounts on desserts for customers who order items from this menu.

Offer club privileges for customers in this segment (e.g., bonus points for purchasing premium dishes or invitations to gastronomic evenings).

Reasons for the choice:

Cluster 4 customers already spend significant amounts on meat and vegetarian meals, so a premium offer will increase their loyalty and average check.

High income allows customers to appreciate and accept more expensive and unique offerings.

Including free compliments or small discounts on desserts will help increase sales in low-spending categories.